



Norman R. Green
norm.green@gemstone.com
Director of Engineering
GemStone Systems Inc.

- GemStone Systems At A Glance (briefly)
 - What Is GemStone/S? (briefly)
 - Why 64 Bit?
- 
- Two decorative geometric patterns are present on the right side of the slide. The upper pattern consists of several small, downward-pointing triangles in shades of yellow, orange, and pink. The lower pattern is a larger, more complex arrangement of triangles in shades of purple, green, and yellow, forming a larger triangular shape.

- Founded 1982
- Headquarters: Beaverton, Oregon, USA
- Privately Hel
- Experienced Engineering Team
 - Many employees with 10+ years tenure
 - Some over 20 years.
- Over 200 installed customers
- 24 x 7 global support.

GemStone/S Customers

Banking / Finance

JPMorganChase 

ICE TRADE THE WORLD™



Government



Office of the
Communications Security
Establishment Commissioner

 **Vitech**
corporation



Government
of Canada

Gouvernement
du Canada

Communications Security Establishment Canada

Telecommunications

BELLSOUTH®

NORTEL
NETWORKS

 Sprint®

 verizon

Transportation/Shipping



We take it personally

 **NYK LINE**
NIPPON Yusen Kaisha







Three Easy Pieces:

1. An Object Oriented Database

- **Objects stored in object format**
 - **No tables,**

- **A**
 - **Atomic – all or nothing**
 - **Consistent – start/end states are consistent**
 - **Isolation – Commit in Session A does not immediately affect Session B**
 - **Durable – cannot be rolled back once committed.**

2. A Smalltalk dialect

- Comes with a complete set of Kernel Classes.

a

rs, etc).

3. An Application Server

- Applications have been written entirely in GemStone/S.
- Java
- C & C++
- UNIX shell (using topaz)

Why GemStone 64 Bit?

- **Customers desired to scale beyond the 32-bit limits:**

- **4 GB Addr**

- **1**



- **Why Do We Care About 32 Bit Limits?**

- **Examples**

- **5 GB reposi**

ed page cache

- **100 GB repository, 2 GB Shared page cache**

- **2% of database cached**

- **Performance: POOR**



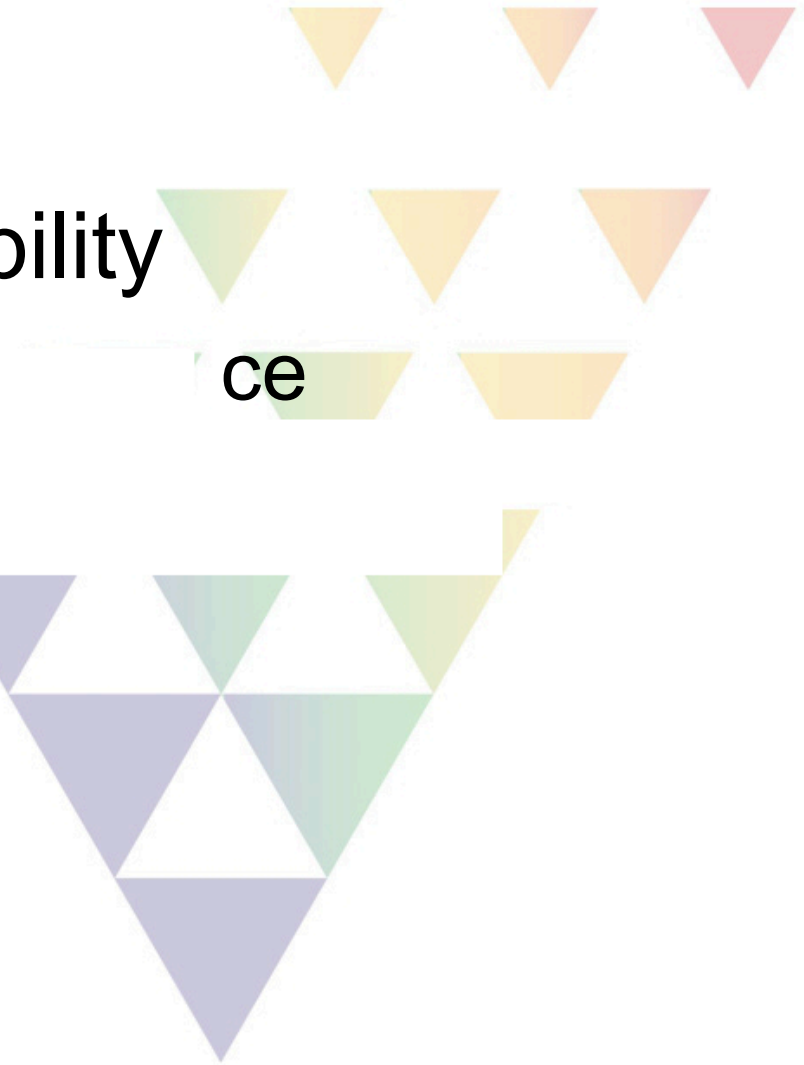
- **Project Divided Into 2 Phases:**
 - **Phase 1: 64 bit address space**
 - **Phase 2: 6 entifiers**

Phase 1 Goals

- Performance Scalability

- Exploit 64

- Re



Phase 1 Features

- 100% 64 bit
- 2 billion O
- Su
 - Solaris 9, 10
 - HP-UX 11.11 (PA-RISC)
 - AIX 5.2, 5.3
 - Linux x86_64 (RH 4, SuSE 9.3)

Phase 1 Features

- GemBuilder for Smalltalk (GBS) Support
 - VisualWor



Phase 1 Improvements

- Supports very large Shared Page Caches



Phase 1 Improvements

➤ Shared Cache Warming

- Loads data

d cache

-

or all data was loaded.

mes full

Phase 1 Improvements

- No more “large object leaks”
 - Old Design
- New Design
 - Objects only go to disk if referenced by a committed object.
 - No exceptions for large objects.
- Net effect: Large reduction in garbage object creation.

Phase 1 Improvements

- Improved Symbol Management
 - Symbol G
 - Guarantees symbols are always canonical
 - Faster Symbol Lookup
 - AllSymbols collection redesigned for speed
 - Lookups now use binary searches

Phase 1 Improvements

➤ Online Backups

- Safely co

nts while system

-

allowed.

borts are

Phase 1 Improvements

➤ Parallelized Garbage Collection

- Reclaim

- Run “online” with production
- Improved reclaim performance for large production systems.

Phase 1 Improvements

- Major Virtual Machine Redesign
 - Copy-on-read (OM)
 - All objects read/written are copied to private VM memory.
 - Large working set = large memory footprint
 - Byte code dispatch loop written in assembler

Phase 1 Improvements

Faster Smalltalk Virtual Machine

Te	GS 6.1	GS64 1.1
100 factorial	2.397	2.092
Commit 6.5 MB data	54.541	20.136
Fault and verify 30 MB data	3.478	2.469
Create & de-ref 6.5MB objs	3.8879	0.3809

Phase 1 Engineering Tasks

- For 615,000 lines of C code:
 - Convert to C
 - long -> int
 - unsigned long -> unsigned int
 - Rewrite object manager and LOM garbage collector from scratch
- Duration: 15 months

Phase 1 Production Customers

- China Ocean Shipping Company (COSCO)
 - Location: Shanghai, China
 - Business: C
-
- Intercontinental Exchange (ICE)
 - Location: Atlanta, GA
 - Business: Energy Futures Trading
 - Platforms: AIX, Windows XP
 - URL: www.theice.com

Phase 1 Production Customers

➤ Northwater Capital

- Location: Toronto, Canada

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➤ Soon: Dutch Agricultural Institute (LEI)

- Location: The Hague, Netherlands
- Business: Government
- Platforms: Solaris, Windows
- URL: www.lei.nl

Phase 2 Goals

- Object Volume Scalability

- Support ve

- Fu

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s

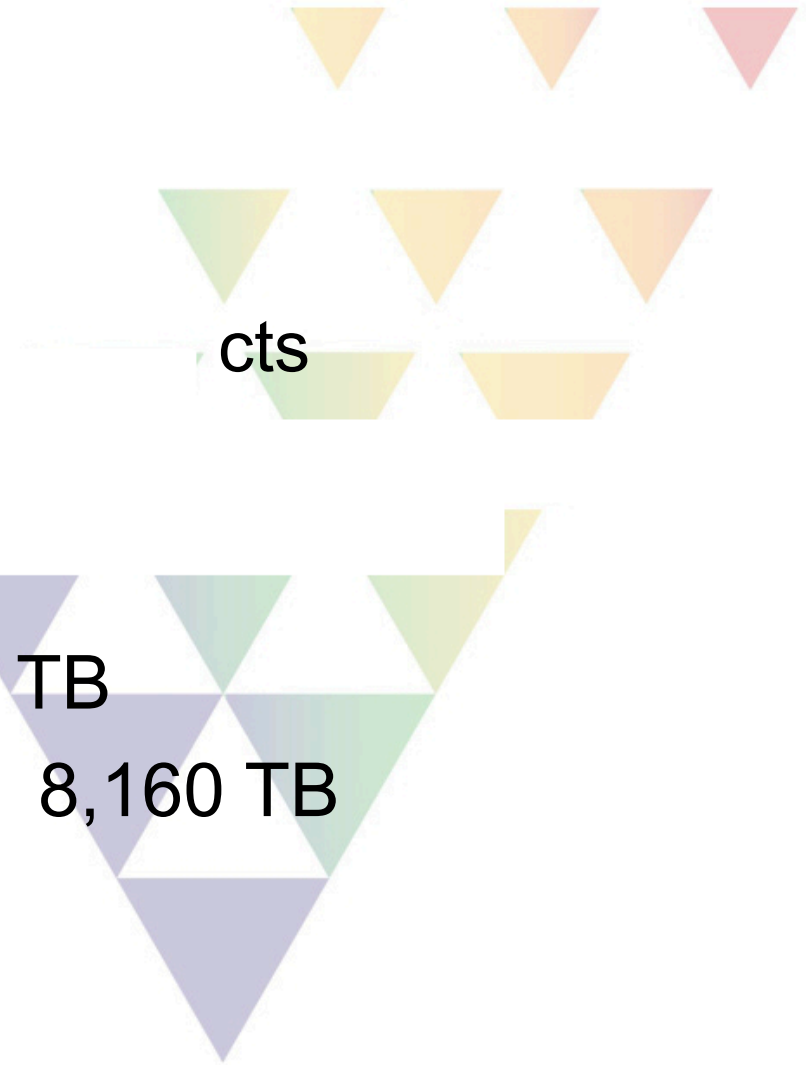
Phase 2 Features

➤ 64 bit object IDs

- Maximum

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- Max extent size: 32 TB
- Max database size: 8,160 TB



Phase 2 Features

➤ Faster Smalltalk VM

- Add 100 A

1k byte codes

➤ Ex

- $\pm 2^{60}$ ($\pm 1,152,921,504,606,846,976$)
- Previous range: $\pm 2^{29}$

Phase 2 Features

- New Special Class: *SmallDouble*
 - Subset of *Double*
 - Range: $5.0e-39$ to $6.0e+38$ (approx)
 - Specials encapsulate their value in the in the object ID
 - No disk I/O

Phase 2 Features

➤ Better Indexes on Collections

- Complete

indexing

-

- Fewer concurrency conflicts
- Same query semantics as before

AllEmployees detect:{:e| e.firstName = 'Stephane' & e.lastName = 'Ducasse' }

GemStone/64 2.x Features

- Conversion From Previous Releases
 - GemSton

GemStone/64 2.0

- Released 3/31/2006
- Supported

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Production Customers

- OOCL (Orient Overseas Container Limited)
 - Bu
 - Platforms: Solaris, HP-UX, Windows XP
 - URL: www.oocl.com

Production Customers

- OOCL

- Production

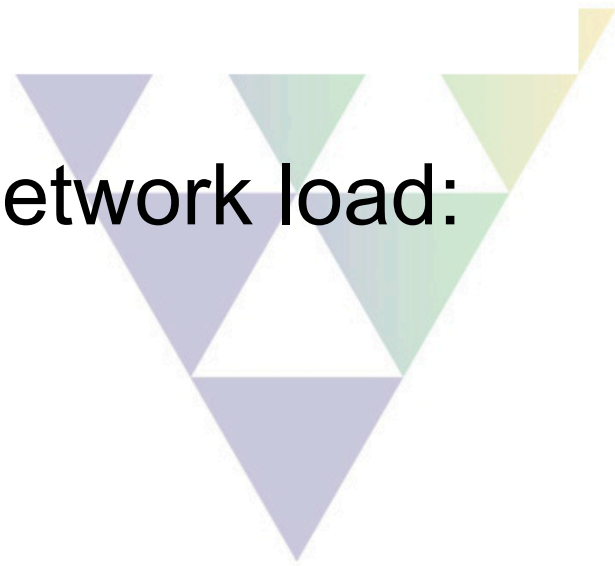
7/30/06

- 1.7 billion objects

- 1800 concurrent users

OOCL

- Performance:
 - Server VM
 - Client – Server Network load:
 - 20% more bytes
 - 32 -> 64 bit oops



OOCL

- Performance:

- VW Client

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- GemBuilder For Smalltalk (GBS) performance loss

- Improved GBS coming in October

- Overall Application Performance: 15% slower.

Integers

GemStone/64 2.1

- Target: October 2006:
- Add suppo

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- VM Performance Improvements
- Index improvements
- Bug fixes

Possible Problems

- In the new GS64 VM design:
 - All objects are interpreted by the interpreter.
- This was not true before GemStone64

Possible Problems: Example

So this code...

`^Object`

...may be fast in GS 6.1, but not in GS 64.

Who Knows Why ?

```
aLargeArray do:[ :each| each == someObject  
  ifTrue:[^each]. ].  
^Object error: #keyNotFound.
```

- == co

GS 64:

- == compares identity, but each must be read into memory.

Solution: Do it this way:

```
|index|  
index := aLargeArr
```

```
cal: someObject.
```

```
ifFals
```

`#indexOfIdentical` **calls a primitive,**
which is smart enough to search
without faulting in the objects in
`aLargeArray`

Other Methods Added To Avoid Unnecessary Object Faulting

- IdentityBag >> copyFrom:count:into:startingAt:
- Ordere
- SequencableCollection >> copyFrom:count:into:startingAt:

- Multi-threaded garbage collection
 - markForCollection, etc
- Multi-threaded
 - Cr
- Multi-thread the Stone process
- Improved VM Performance

- Additional Special Classes
 - Candidates:
 - Date
 - ScaledDecimal
- Reduced Conflict (RC) Indexes on Collections

- Thread-safe C-interface (GCI layer)
- Support for Seaside
 - Need native

inuations

Questions?

